Exercise. Sinusodial Transverse Wave

Wave Characteristics:

- $\lambda = 20\,cm$ ∄ motion (+)x direction
- positive displacement (y)
- at $x = 0$ ∄ $y = 0$, $t = 0$
- $T = 10\,s$
- Amplitude, $y_m = \pm 4\,cm$

a) At $t = 0$ is the plot y vs x a (+) or (−) sine function?

The graph $y = y_m \sin(-\omega t + \phi)$ is → positive if $\phi = 0, 2\pi, 4\pi \ldots$

↳ negative if $\phi = \pi, 3\pi, 5\pi \ldots$

b) Find the wave function $y(x,t)$.

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.2m} = \underline{10\pi \text{ rad/m}}$$

$$T = 10s \Rightarrow \omega = \frac{2\pi}{T} = 0.2\pi \ \underline{\text{rad/s}}$$

$\phi = \pi, 3\pi, \ldots \rightarrow$ Set $\phi = \pi$

Sign $-\omega \rightarrow$ travels in the x-direction

$$v = \lambda f = (0.2m)(1/10s) = \underline{0.02 \text{ m/s}}$$

$$\therefore y(x,t) = 0.04 \sin(10\pi x - 0.2\pi t + \pi)$$

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c) Find $y(x=0,\ t=5.0s)$

$y(0,5) = 0$

d) Find particle velocity at

$x=0,\ t=5.0s$

$$u = \frac{\partial y(x,t)}{\partial t} = y_m \omega \cdot \cos(kx - \omega t + \phi)$$

$\phi = \pi$

$$\therefore u = (0.04)(0.2\pi)\cos(0 - \pi + \pi)$$

$$= -(8\times10^{-3})\pi \text{ m/s}$$